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Numerical Heat Transfer, Part A: Applications
An International Journal of Computation and Methodology

ISSN: 1040-7782 (Print) 1521-0634 (Online) Journal homepage: http://www.tandfonline.com/loi/unht20

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To cite this article: Anoop K. Gupta, Pragya Mishra & R. P. Chhabra (2017): Momentum and heat transfer characteristics of a thin circular disk in Bingham plastic fluids, Numerical Heat Transfer, Part A: Applications, DOI: 10.1080/10407782.2017.1412219

To link to this article: https://doi.org/10.1080/10407782.2017.1412219

Published online: 21 Dec 2017.

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NUMERICAL HEAT TRANSFER, PART A
https://doi.org/10.1080/10407782.2017.1412219

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Momentum and heat transfer characteristics of a thin circular disk in Bingham plastic fluids

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ABSTRACT

The laminar forced convection momentum and heat transfer aspects of a circular disk oriented normal to the flow and maintained at a constant flux or a constant temperature condition in a stream of a Bingham plastic fluid are studied over wide ranges of parameters as follows: Reynolds number, $Re \leq 150$; Prandtl number, $1 \leq Pr \leq 100$; Bingham number, $Bn \leq 100$, and thickness-to-diameter ratio, $0.01 \leq (t/d) \leq 0.075$. The new results on hydrodynamics are analyzed in terms of streamline plots, recirculation length, morphology of yielded/unyielded regions, and drag coefficient, and on heat transfer aspects in terms of isotherm contours, local and average Nusselt number. The flow domain is spanned by the simultaneous existence of the yielded and unyielded sub-regions, depending upon the relative strengths of the fluid inertia (Re) and yield stress (Bn). All else being equal, the rate of heat transfer is higher for an isothermal disk than that for the isoflux condition. Both the drag and average Nusselt number bear a positive dependence on the Bingham number. The drag is influenced only slightly (~5%) by thickness (t/d); however, the heat transfer can increase on this count by up to 15% under appropriate conditions. Finally, the present numerical results on drag and Nusselt number (in terms of j_H-factor) have been correlated via simple empirical equations using the modified definitions of the Reynolds (Re^*) and Prandtl number (Pr^*), thereby enabling *a priori* estimation of drag and heat transfer in a new application.

ARTICLE HISTORY
Received 7 September 2017
Accepted 14 November 2017

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1. Introduction

The flow of fluids over objects of three-dimensional shapes such as a sphere, spheroid, cone, disk, etc. denotes an important class of boundary layer problems in the domain of fluid mechanics due to their fundamental and practical significance. Typical examples include the flow in fixed- and fluidized beds employing catalysts of various shapes, gravity settlers, mixing vessels, processing of granular materials, coal combustion, etc [1–4]. Notwithstanding the fact that most of the aforementioned situations entail multiple particles, the experience has shown that a satisfactory understanding of the behavior of a single particle is a necessary prerequisite to understand the behavior of real systems. For instance, the hydrodynamic drag on a single particle is amended to estimate the drag on an ensemble of particles. Conversely, the terminal velocity of a single particle is often used as a guide to design and operate slurry pipelines and fluidized bed reactors, for example, see [5, 6] etc. Similarly, from a theoretical standpoint, such model shapes serve as a launching pad to understand the physics of more complex flows involving multi-body and body–wall interactions such as that encountered in the flow of suspensions. Furthermore, it is often needed to estimate the falling velocity of a particle in a fluid or the force needed to move a particle in a fluid at a desired speed. This information is

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Nomenclature

A_p	projected area of the disk normal to the direction of flow (m^2)
Bn	Bingham number $(= \tau_0 d/\mu_B V_0)$ (dimensionless)
Bn^*	modified Bingham number $(= Bn/(1 + Bn))$ (dimensionless)
Bn_{cr}	critical Bingham number for cessation of flow separation (dimensionless)
Br	Brinkman number $(= \mu_B V_0^2/k(T_w - T_\infty))$ (dimensionless)
C	thermal heat capacity of fluid (J kg^{-1} K^{-1})
C_D	total drag coefficient $(= 2F_D/\rho V_0^2 A_p)$ (dimensionless)
C_D^*	drag coefficient $(= F_D/\tau_0 \cdot \pi R^2)$ (dimensionless)
C_{DF}	friction (or viscous) drag coefficient $(= 2F_{DF}/\rho V_0^2 A_p)$ (dimensionless)
C_{DP}	pressure (or form) drag coefficient $(= 2F_{DP}/\rho V_0^2 A_p)$ (dimensionless)
C_{DS}	Stokes drag coefficient (dimensionless)
C_p	pressure coefficient $(= 2(p'_w - p_{ref})/\rho V_0^2)$ (dimensionless)
d	disk diameter (m)
F_D	drag force (N)
F_{DF}	friction (viscous) component of the drag force (N)
F_{DP}	pressure (form) component of the drag force (N)
h	local heat transfer coefficient (W m^{-2} K^{-1})
j_H	Colburn j_H-factor for heat transfer $(= Nu/Re^* \cdot Pr^{*1/3})$ (dimensionless)
k	thermal conductivity of fluid (W m^{-1} K^{-1})
L	size of the computational domain (m)
L_r	recirculation length measured from the rear stagnation point (dimensionless)
M	parameter used in Papanastasiou model, Eq. (8) (dimensionless)
n_s	unit vector normal to the surface of disk (dimensionless)
N	total number of cells in the half computational domain (dimensionless)
N_p	number of grid points on the surface of disk (dimensionless)
Nu	average Nusselt number (dimensionless)
Nu_H	average Nusselt number for CHF case (dimensionless)
Nu_l	local Nusselt number on the surface of disk (dimensionless)
Nu_T	average Nusselt number for CWT case (dimensionless)
Nu_∞	conduction limit of the Nusselt number (dimensionless)
p	pressure (dimensionless)
p_{ref}	reference pressure (Pa)
Pe	Peclet number $(= Re \cdot Pr)$ (dimensionless)
Pr	Prandtl number (dimensionless)
Pr^*	modified Prandtl number $(= Pr\,(1 + Bn))$ (dimensionless)
q_0	uniform heat flux on the surface of the disk (W m^{-2})
r, z	cylindrical coordinates
R	radius of the disk (m)
Re	Reynolds number $(= dV_0\rho/\mu_B)$ (dimensionless)
Re_{cr}	critical Reynolds number for the onset of flow separation (dimensionless)
Re^*	modified Reynolds number $(= Re/(1 + Bn))$ (dimensionless)
S	surface area of disk (m^2)
t	disk thickness (m)
T	temperature of fluid (K)
V	velocity vector (dimensionless)
V_0	free stream velocity (m s^{-1})
V_r, V_z	velocity components in r- and z-directions (dimensionless)

Greek symbols

∇	del operator (dimensionless)
$\dot{\gamma}$	rate of deformation tensor (dimensionless)
Δ	minimum grid spacing on the surface of disk (m)
η	scalar viscosity function (dimensionless)
λ	parameter used in Bercovier and Engelman model, Eq. (10) (dimensionless)
μ_B	fluid viscosity (Pa s)
μ_y	yielding viscosity used in bi-viscosity model, Eq. (9) (Pa s)
ξ	temperature of fluid (dimensionless)
ρ	density of fluid (kg m^3)
τ	deviatoric stress tensor (dimensionless)
τ_0	fluid yield stress (Pa)

Subscripts

$'$	dimensional variable
0	corresponds to faraway (free stream) condition
w	condition at disk surface

Abbreviations

CHF	constant heat flux
CWT	constant wall temperature

frequently employed in the design of liquid–solid processing equipment such as hydro-cyclones, slurry pipelines, centrifuges, etc. Similarly, such products are also required to be heated or cooled due to their satisfactory end use or other processing requirements [7]. A cursory inspection of the available literature in this area reveals that most of such studies are limited to spheres and sections thereof, and spheroids, while very little is known about the flow and heat transfer aspects of a circular disk, even in Newtonian fluids. This shape is encountered in the cooling of electronic components

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such as disk-type resistors and power transistors for temperature and air flow measurements, disk-type thermistors, silicon wafers, shell-and-tube heat exchangers with disk-and-doughnut baffles [8], parallel plate viscometers, and thermal treatment of food- and pharmaceutical products, are among the few instances where the disk-like particles are encountered. Further examples are found in biomedical and biological engineering applications such as in the context of blood flow.

The first attempt to evaluate the drag on a thin disk in Newtonian liquids was made by Overbeck [9] for oblate spheroids which approaches a disk in the limit of zero thickness. He obtained a closed form expression for drag in the creeping flow limit. Other early studies have explored wall effects on disks settling in a cylindrical tube or towards a plane wall [10–12]. The weak inertial effects ($Re < 5$) for an unconfined flow were considered by Pulley et al. [13]. The range of these results has been subsequently extended up to $Re \leq 100$ by Michael [14] and Munshi et al. [15]. Most of the pertinent studies have been critically reviewed by Munshi et al. [15] and subsequently by Nitin and Chhabra [16, 17] who also investigated the flow past a thin disk in unconfined [15] and confined conditions [16, 17] for a range of blockage ratios (disk diameter/cylinder diameter = 0.02 to 0.5) and disk thickness (t/d) = 0.01 to 0.075. All in all, the drag coefficient was found to increase with the increasing blockage ratio, especially at small Reynolds numbers, similar to the behavior seen for a sphere [3]. Also, the value of t/d in the range 0.01–0.025 was found to be a good approximation for the drag values to be free from disk thickness effects; besides, the influence of disk thickness progressively disappears as the Reynolds number is gradually increased because the drag is predominantly made up of the form drag under these conditions. A complete numerical study of the flow past a disk ($t/d = 0.1$) was conducted by Shenoy and Kleinstreuer [18] in the range $10 \leq Re \leq 300$. They showed that the flow remains axisymmetric and steady up to $Re < 135$ in Newtonian fluids and loses its planar symmetry in the range $135 \leq Re < 155$ and finally, it becomes three-dimensional for $Re \geq 155$. However, the major thrust of this study was on the delineation of the onset of instabilities and of the different flow regimes which were seen to be fundamentally different from that reported for a sphere. Similarly, based on their experimental results for a freely falling disk, Zhong and Lee [19] concluded that the azimuthal symmetry of the flow about the central axis is lost at $Re = 105$ while the flow becomes unsteady for $Re > 118$. Finally, the periodic vortex shedding commences at $Re > 150$. The agreement between the numerical and experimental drag values is satisfactory up to about $Re = 100$. The preceding numerical activity has also been supplemented by limited experimental studies in this field as far as the blockage effects and drag in the steady axisymmetric regime are concerned [20–22]. On the other hand, Zhong et al. [23] have experimentally investigated different flow regimes (steady, planar zigzag, spiral, tumbling, transitional, chaotic, etc.) for a freely falling thin disk in Newtonian fluids using the *stereoscopic vision method*. Similarly, based on their numerical results, Chrust et al. [24] have described various transitions observed for a freely falling thin disk in Newtonian fluids in terms of the dimensionless Galileo number and the moment of inertia of the disk; the steady axisymmetric flow regime being the simplest in which the moment of inertia of the falling disk has no effect at all. In this regime, the hydrodynamic drag on the disk just balances the buoyant weight of the disk. Their predictions of the other flow regimes such as fluttering, zig–zag, spiral motion, etc., are also in line with the experimental observations documented in the literature [19, 23, 25]. However, such flow regimes are generally observed with disks of low moment of inertia and/or in thin (less viscous) fluids like air (e.g., leaves falling in air). The corresponding heat transfer results for disks are even scarcer than the drag studies. For instance, Kobus and Wedekind [26], Wedekind and Kobus [27], and Kobus and Shumway [28] have presented experimental Nusselt number data in the forced-, free-, and mixed-convection (both assisting and opposing flow) regimes for varying disk thicknesses as $0.062 \leq t/d \leq 0.2$. They developed dimensionless correlations over wide ranges of Reynolds number ($900 \leq Re \leq 3 \times 10^4$) and Rayleigh number ($\leq 10^5$) to consolidate their experimental results. Wedekind [29] performed experiments on disk-type thermistors to study the laminar forced-convection from heated circular disks at $Pr = 1$. These studies [26–29] suggest that the heat transfer results for thin circular disks are similar to that of a flat plate and the data for both geometrical configurations can be collapsed using an equivalent flat plate with an effective length for the disk which is the average

distance that a fluid particle travels in the boundary layer formed on the disk. In addition to this, few numerical studies are also available on natural convection from disks in Newtonian fluids [30–32] using series expansion method [30, 31] or thin boundary layer approximation [32]. For instance, Merkin [30, 31] showed that the boundary-layer thickness was very large near the center of the disk and thus, the flow splits into two distinct regions, namely, a thin viscous region close to the disk of thickness of $O(r^{2/3})$, and a thick outer inviscid region of thickness of $O(r^{-2/3})$. Robinson and Liburdy [32] casted their results in terms of the radial position and Grashof number and concluded that beyond the Rayleigh number of about 10^6, the outer flow field has virtually no effect on the rate of heat transfer. Thus, all in all, a reasonable body of information is now available on the drag and heat transfer aspects of thin disks in Newtonian fluids, albeit most of these relate to air as the working fluid, that is, for a constant value of $Pr = 0.7$ and thus little is known about the effect of Prandtl number on heat transfer even in Newtonian fluids.

In contrast, it is readily acknowledged that most structured and high molecular weight polymer suspensions, foams, emulsions, pastes, and particulate slurries encountered in a broad spectrum of industrial applications are characterized by the existence of a yield stress which must be overcome for such a material to deform like a fluid [33–35]. Thus, depending upon the magnitude of the local stress level in the flow field, such a substance may behave like an elastic solid (unyielded) or like a viscous fluid (yielded). This family of materials is known as viscoplastic and their behavior is typically approximated by the familiar Bingham plastic fluid model. Consequently, in a given flow field, both solid-like and fluid-like regions co-exist; their size, shape, and location depend on the flow conditions in the vicinity of the object. Over the past 25–30 years, much research effort has been directed at understanding the flow of viscoplastic fluids past variously shaped three-dimensional objects. Undoubtedly, extensive literature now exists on the flow of Bingham plastic and Herschel Bulkley model fluids past a sphere [3]. Since bulk of the literature up to ~2006 has been reviewed in [3], only the salient points are recounted here. First, good correspondence between the numerical predictions and experimental values of drag for a sphere up to the Reynolds number of ~100 exists, thereby lending credibility to the results available in this field [3, 36–42]. Also, the effect of Bingham number progressively diminishes with the increasing Reynolds number. The corresponding heat transfer results in the laminar forced- [41] and free-convection [43] regimes are of relatively recent vintage. In free convection, the mean Nusselt number for a sphere decreases from its maximum value (at $Bn = 0$, i.e., in Newtonian fluids) to the expected conduction limit of $Nu_\infty = 2$ at large Bingham numbers where most of the fluid is unyielded and thus heat transfer occurs mainly by conduction. In contrast, in the forced convection regime, the Nusselt number bears a positive dependence on the Bingham number due to the gradual sharpening of the temperature gradient on the sphere surface. However, no experimental results are available on heat transfer from a sphere in viscoplastic fluids to corroborate the numerical predictions of Nusselt number. Limited results are also available on the effect of blockage on the drag and heat transfer of a sphere undergoing steady translation in Bingham plastic fluids [44, 45]. These results clearly show that the falling sphere sees the wall, so to say, only if the fluid-like region intersects with the physical wall. The other three-dimensional shapes which have been studied include spheroids [46, 47] and hemispheres [48–50]. Qualitatively similar trends are obtained as far as the scaling of the drag and Nusselt number with the Reynolds number, Prandtl number, and Bingham number is concerned. The aforementioned numerical effort in this field has also been supported by extensive experimental measurement of drag on variously shaped objects including cones, thin disks, and plates [3, 22, 51–57]. However, only limited comparisons between the predictions and experimental results have been carried out and these are encouraging, for example, see [22] for Newtonian and power-law fluids for circular disks and square plates. Such comparisons for Bingham plastic fluids have proved to be difficult for two reasons: in some cases, the values of the Bingham yield stress used in experiments are in doubt due to its evaluation by extrapolation of the rheological data to zero shear rate value. Second, in many instances, the range of Bingham and Reynolds numbers is way beyond the limits of the currently available numerical predictions. It is thus abundantly clear that no prior results, barring the scant experimental data of Boardman and

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Whitmore [51], Brookes and Whitmore [52], Pazwash [53], and Pazwash and Robertson [54] on drag, are available on the momentum and heat transfer aspects of a thin circular disk in yield stress fluids. This knowledge gap is addressed in this study. However, before leaving this section, it is appropriate to add here that the hydrodynamics and heat transfer characteristics of a rotating disk (of uniform and variable thickness) have also been investigated extensively, in Newtonian, power-law, and Bingham plastic fluids, for example, see [58–63], using a range of analysis tools including boundary layer flow approximation, homotopy analysis, similarity transformation, etc. To exploit the similarity transformation approach, the steady flow equations necessitate the disk thickness to vary in a specific fashion. Such studies have been motivated, in part, as a possible tool to develop the so-called rotating disk viscometer and/or to measure molecular diffusivity. Thus, for instance, Xun et al. [61] have identified a suitable similarity variable for power-law fluids to convert the partial differential equations into a set of semi-similarity ordinary differential equation. Griffiths [63] has considered the boundary layer flow of power-law, Bingham plastic, and Carreau model fluids over a rotating disk. They reported extensive results on skin friction and Nusselt number.

In this work, extensive numerical results on the drag and Nusselt number are reported for a thin circular disk of varying thickness $0.01 \leq (t/d) \leq 0.075$ over the following ranges of kinematic conditions: $Re \leq 150$; $1 \leq Pr \leq 100$ and $0 \leq Bn \leq 100$. Furthermore, the influence of the two commonly employed thermal boundary conditions, that is, isothermal (constant wall temperature (CWT)) or isoflux (constant heat flux (CHF)), prescribed on the surface of the disk is also studied in terms of the overall heat transfer characteristics. Additional insights are developed by analyzing the streamline plots, morphology of yielded/unyielded regions, isotherm patterns, and local Nusselt number in the proximity of the disk.

2. Problem formulation and governing equations

Consider a thin circular disk of diameter d and thickness t placed transverse to the flow of a Bingham plastic fluid (V_0, T_0), as shown schematically in Figure 1*a*. The surface of the circular disk is maintained either at a CHF of q_0 W/m^2 (CHF condition) or at a constant temperature of T_w (CWT condition). To approach the truly unconfined flow situation, the disk is positioned at the center of a tubular domain of length and diameter of $2L$, as shown in Figure 1*b* along with the boundary conditions (dimensionless) used here. The flow is assumed to be axisymmetric, steady, incompressible, and laminar. The thermophysical properties of the fluid are assumed to be temperature-invariant over the narrow range of temperature difference prescribed here. Finally, over the range of conditions spanned here, the maximum value of the Brinkman number (Br) is of the order of $\sim 10^{-3}$ and thus it is fair to neglect the viscous dissipation effects in the energy equation. Under these conditions, the governing differential equations are written in their dimensionless forms as follows:

$$\text{Continuity equation}: \ \nabla . V = 0 \tag{1}$$

$$\text{Momentum equation}: \ (V \cdot \nabla) V = -\nabla p + \frac{(\nabla \cdot \tau)}{Re} \tag{2}$$

$$\text{Thermal energy equation}: \ (V \cdot \nabla)\xi = \frac{(\nabla^2 \xi)}{Re \cdot Pr} \tag{3}$$

For an incompressible fluid, the extra stress tensor (τ) is expressed in terms of the rate of strain tensor ($\dot{\gamma}$) as:

$$\tau = \eta \dot{\gamma} \tag{4}$$

The magnitudes of the extra stress and rate of strain tensors are given as:

$$|\tau| = \sqrt{tr(\tau^2)} \tag{5}$$

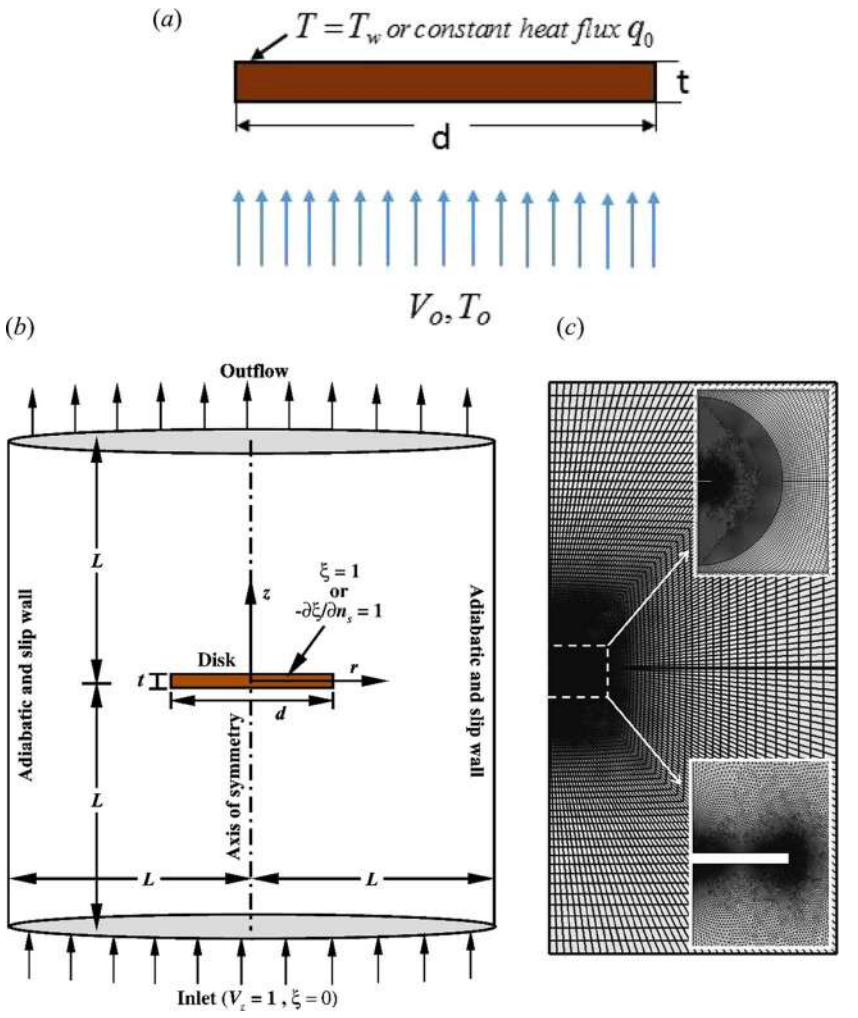


Figure 1. (*a*) Schematics of the flow, (*b*) computational domain along with the boundary conditions (dimensionless), and (*c*) typical grid structure.

$$|\dot{\gamma}| = \sqrt{tr(\dot{\gamma}^2)} \tag{6}$$

For a Bingham plastic fluid, the rheological equation of state (dimensionless) is expressed as:

$$\dot{\gamma} = 0, \text{ if } |\tau| \leq Bn \tag{7a}$$

$$\tau = \left(1 + \frac{Bn}{|\dot{\gamma}|}\right)\dot{\gamma}, \text{ if } |\tau| > Bn \tag{7b}$$

Naturally, the discontinuous form of Eq. (7) renders it difficult to incorporate it directly into a numerical solution scheme. Consequently, over the years, several smoothening or regularization approaches have been developed which convert the abrupt transition between the yielded and unyielded regions into a gradual one [64]. Over the years, three such approaches, namely, exponential due to Papanastasiou [65], bi-viscous [66], and Bercovier and Engleman [67] have been shown to yield comparable results for a range of flows as far as the gross engineering parameters such as drag coefficient and Nusselt number, etc., are concerned provided suitable values of the regularization

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parameters are used. However, none of these approximations resolve the stress field entirely satisfactorily. In recent years, there has been a growing interest in developing tools which obviate the need for such regularization approximations including *operator-splitting method* [68, 69], *Augmented Lagrangian method* [70], the so-called *damped Newton method* [71] or the one based on the variational inequalities [64]. Undoubtedly, the latter methods perhaps resolve the velocity and stress fields much more precisely than the regularization methods, these are also much more computer resource intensive. Furthermore, in a few instances like for the creeping motion of a rigid sphere and a gas bubble and for free-convection in a square cavity, detailed comparisons between the results obtained with and without the regularization methods show a close correspondence between the two predictions, albeit the differences between the two results increase with the increasing Bingham number. This transpires confidence in employing the regularized form of Eq. (7) in this work. While bulk of the results included here are based on the exponential smoothening function [65], limited results have also been obtained using the other two approaches [66, 67] to demonstrate that the present results are free from such artifacts. Thus, within the framework of the above-mentioned three regularization methods, the scalar viscosity function, (η) can be written (in its dimensionless form) as:

a. Papanastasiou model [65]:

$$\eta = 1 + \frac{\left(Bn\left\{1 - \exp\left(-M\sqrt{tr(\dot{\gamma}^2)}\right)\right\}\right)}{\sqrt{tr(\dot{\gamma}^2)}} \tag{8}$$

b. Bi-viscosity model [66]:

$$\eta = (\mu_y/\mu_B), \text{ for } |\tau| \le Bn \tag{9a}$$

$$\eta = 1 + \frac{Bn}{\sqrt{tr(\dot{\gamma}^2)}}\left(1 - \frac{1}{\mu_y/\mu_B}\right), \text{ for } |\tau| > Bn \tag{9b}$$

c. Bercovier and Engelman model [67]:

$$\eta = 1 + Bn\left(\frac{1}{\left(\sqrt{tr(\dot{\gamma}^2)} + \lambda\right)}\right) \tag{10}$$

Equations (8) and (9) effectively replace the unyielded material by a hypothetical fluid of very high viscosity ($>>> \mu_B$). On the other hand, Eq. (10) allows extremely small strain rates in the solid-like unyielded domains as opposed to its zero value expected in an ideal elastic material.

Lastly, the following boundary conditions have been applied here:

- *At the inlet*: $V_r = 0,\ V_z = 1,\ \xi = 0.$
- *At the outlet:* $\partial\varnothing/\partial z = 0$ where ($\varnothing = V_r,\ V_z$, or ξ).
- *On the surface of disk:* $V_r = V_z = 0$ and $\xi = 1$ (CWT) or $-\partial\xi/\partial n_s = 1$ (CHF).
- *At the slip (wall) boundary*: $\partial V_z/\partial r = 0,\ V_r = 0,\ \xi = 0.$

Since over the range of conditions spanned here, the flow is axisymmetric, the computations have been carried out only in the half domain. The preceding equations have been non-dimensionalized using d and V_0 as the characteristic linear and velocity scales. The consistent scales are ρV_0^2 for pressure, $\mu_B V_0/d$ for stress, and V_0/d for shear rate components. The non-dimensional temperature is given by $\xi = (T - T_0)/(T_w - T_0)$ and $\xi = (T - T_0)/(q_0 d/k)$ for the CWT and CHF thermal boundary conditions, respectively. Evidently, the velocity and temperature fields here are expected to be functions of four dimensionless groups, namely, Reynolds number (Re), Prandtl number (Pr), Bingham number (Bn), and thickness-to-diameter ratio (t/d) for a given thermal boundary

condition. Finally, the numerical solution of Eqs. (1)–(10) along with the aforementioned boundary conditions maps the flow domain in terms of the primitive variables (V, p, ξ) which, in turn, are post-processed to obtain the derived quantities like drag coefficient and Nusselt number as detailed in our previous studies [41, 45–50]. It is also worthwhile to add here that the use of μ_B as the characteristic viscosity does not incorporate the role of fluid yield stress. Therefore, several authors [41, 46, 48–50, 72] have advocated the use of the modified definitions of the Reynolds (Re^*) and Prandtl (Pr^*) numbers based on the effective viscosity given by $\mu_B + \tau_0/(V_0/d)$ to account for the effect of fluid yield stress (Bn) as:

$$Re^* = \frac{Re}{(1+Bn)} \text{ and } Pr^* = Pr(1+Bn) \tag{11}$$

Based on these modified definitions of the Reynolds and Prandtl numbers, the following functional relationships can be postulated for the drag and Nusselt number:

$$C_D = f(Re^*,\ t/d) \tag{12}$$

$$Nu = f(Re^*, Pr^*,\ t/d) \tag{13}$$

This work endeavors to explore and develop these relationships. It is worthwhile to note here that, in the limit of the Newtonian fluid behavior ($Bn = 0$), the two definitions of the Reynolds and Prandtl numbers coincide. Perhaps, the use of Re^* and Pr^* as opposed to that of Re, Pr, and Bn provides a fair basis for comparison between the results for Newtonian and Bingham fluids. Finally, the application of the classical dimensional analysis will yield Re, Bn, and Pr as the pertinent dimensionless numbers, whereas the formulation of Re^* and Pr^* entails the additional physical insights by incorporating the contribution of the fluid yield stress in estimating the effective fluid viscosity. It is thus preferred in situations dealing with viscoplastic fluids, as has been argued by Thompson and Soares [72] recently. In the present work, both these approaches have been used.

3. Numerical solution methodology and computational parameters

In this work, the governing differential equations (1)–(3), along with the boundary conditions, have been solved numerically using the finite-element-based COMSOL Multiphysics® (version 4.3a). Since the solution methodology used here is similar to that implemented in some of our recent studies [41, 45–50], only the key points are recapitulated here. The *steady, axisymmetric,* and *laminar flow* module was used along with the physics of *heat transfer in fluids*. The velocity (second order) and pressure (linear) fields were discretized using the Lagrangian P_2–P_1 scheme. The **PAR**allel Sparse Linear **DI**rect **SO**lver was employed to solve the system of equations. This solver discretizes the governing equations by the finite element method resulting in a sparse linear system. The extra stress tensor in the momentum equation was approximated using the regularized Bingham model via a user-defined function. Finally, the following relative convergence criterion was used for all primitive variables (pressure, velocity, and temperature) which also ensured that the drag and Nusselt number values had stabilized at least up to four significant digits:

$$\left|\frac{\Gamma^{n+1} - \Gamma^n}{\Gamma^{n+1}}\right| \leq 10^{-6} \tag{14}$$

where n and Γ are the iteration number and independent variables such as p, V, and ξ.

The yield surfaces separating the fluid-like and solid-like sub-regions were delineated using the von Mises yield criterion [73] with a relative tolerance of 10^{-6}. This implies that the material was assumed to be unyielded when the percentage difference between the magnitude of the local stress tensor at a point and the fluid yield stress was $\leq$0.0001%.

Based on an extensive numerical exploration similar to that used in our recent studies [41, 45–50], the values of M, μ_γ, λ, size of the tubular computational domain (value of L), and a suitable

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computational mesh were finally selected. The domain size of $L/d = 50$ was found to be sufficient for present results to be free from boundary effects. To capture steep velocity and temperature gradients close to the disk, the flow domain was sub-divided into two sub-regions; the inner region extends up to a diameter of $\sim 6d$ and the outer region spans the rest of the region as shown in Figure 1*c*. The inner region was meshed with very fine triangular elements, whereas non-uniform quadrilateral grids were used in the outer region to economize on the computational effort without compromising the precision of the present results. Also, relatively fine elements were located in the vicinity of the interface of yielded/unyielded regions to adequately delineate the yield surfaces. Three different grids were created for different values of disk thicknesses, that is, ($t/d = 0.01$, 0.05, and 0.075) and their effects on the drag and Nusselt number at the extreme values of the Reynolds number, Bingham number, and Prandtl number, that is, $Re = 150$, $Bn = 100$, and $Pr = 100$ were examined. Finally, grids (comprising $N = 71{,}343$, 60,432, and 61,145 CVs in half computational domain for $t/d = 0.01$, 0.05, and 0.075, respectively) were found to be adequate in each case in the present study. This corresponds to the number of degrees of freedom ~1,76,000 for all cases. Finally, the true Bingham fluid viscosity was approximated with a suitable value of the regularization parameter, M which was justified from the predictions of the yield surfaces (not shown here for the sake of brevity) and the values of the drag and Nusselt number at $Pr = 100$ and $Bn = 100$. The results had also stabilized for all values of $M \geq 10^7$ at least up to four significant digits. Limited tests were also performed using the bi-viscosity and Bercovier and Engelman techniques and these results were found to change negligibly beyond $\mu_y/\mu_B \geq 10^6$ or below $\lambda \leq 10^{-8}$. Indeed, the resulting values of the macroscopic parameters, namely, C_D, C_{DP}, and *Nu,* based on the three regularization schemes were virtually indistinguishable from each other, thereby lending credibility to the reliability of the results reported herein. Thus, in summary, for all values of the disk thickness (t/d), the results reported herein are based on the following selection of the numerical parameters: $L/d = 50$, grid comprising of 71,343, 60,432, and 61,145 nodes and $M \geq 10^7$ or $\mu_y/\mu_B \geq 10^6$ or $\lambda \leq 10^{-8}$.

4. Results and discussion

The structure of the flow field is examined in terms of the streamline patterns, yielded/unyielded zones, isotherm contours, and the distribution of the local Nusselt number along the body contour. The overall behavior is captured in terms of the drag coefficient and average Nusselt number. While most of the results reported herein relate to the disk thickness of $t/d = 0.05$, the influence of the disk thickness on the overall drag and Nusselt number is also demonstrated by varying the value of t/d from 0.01 to 0.075. This range is also in line with the experimental results available in the literature for Newtonian [22, 74], power-law [22], and Bingham plastic [56, 57] fluids. Thus, for instance, the experimental results of Ui et al. [76] on disks falling slowly ($Re \to 0$) in Newtonian fluids span the range $0 \leq (t/d) \leq \sim 1$; the corresponding ranges in the other experimental studies are $\sim 0.02 \leq (t/d) \leq 0.17$ [22] and $0.053 \leq (t/d) \leq 0.16$ [56]. Thus, the seven-fold range of the value of (t/d) covered here is regarded to be sufficient to delineate the influence of this parameter. Similarly, a comment regarding the choice of the parametric ranges used in this work is in order here. Since this work deals with Bingham fluids, no gas exhibits this fluid behavior. The viscosity of a typical Newtonian liquid is known to be at least 100 times that of a gas and therefore, the resulting Reynolds numbers are correspondingly much smaller. On this count, the Bingham plastic fluids are known to be even more viscous [3, 33] and thus the Reynolds number values are typically of the order of 100–150. This coupled with the fact that the fluid yield stress imparts stability, the flow is assumed to be axisymmetric here up to the maximum value of $Re = 150$ and the corresponding values of $Re^* < 150$. It is useful to recall here that the flow past a circular disk is known to be steady and axisymmetric in Newtonian fluids up to $Re = 135$ [18–20, 23]. Thus, in view of the additional viscous effects stemming from the fluid yield stress, the assumption of the steady and axisymmetric flow in this work up to $Re = 150$ or $Re^* = 150$ is justified. Similarly, many fluids of practical interest such as light organics, oils, ethylene glycol, glycerol, etc. exhibit the value of the Prandtl number as

large as 1000 or even higher. However, ultrafine computational meshes are required to resolve very thin thermal boundary layers ($\delta \sim Pr^{-1/3}$) under these conditions. On the other hand, a 100-fold range of Prandtl number ($1 \leq Pr \leq 100$) is regarded to be sufficient to establish the scaling of the Nusselt number with Prandtl number. Finally, the upper limit of the Bingham number, $Bn = 100$ is also chosen due to the fact that this value of Bn corresponds to the condition that yield stress effects are of the order of 100 times the flow stresses ($\tau_0 \sim 100\mu_B V_0/d$) and thus it enables to contrast the results for a Newtonian fluid ($\tau_0 = 0$) with a Bingham fluid over a wide range of yield stress and thus, the range of conditions employed here have been guided by a combination of physical, practical, and numerical considerations. However, prior to undertaking the detailed discussion of the new results, it is important to ascertain the accuracy and reliability of the present results by presenting a series of benchmark comparisons with the relevant experimental results in the next section.

4.1. Validation of results

Much of the relevant literature on the hydrodynamic of circular disks in Newtonian fluids is reviewed, among others, by Clift et al. [20] and Chhabra et al. [21, 22]. Based on the available analytical and experimental results in Newtonian fluids, Clift et al. [20] recommend the following drag expression for a thin disk falling with its flat surface transverse to the direction of flow:

$$\text{For } Re \leq 1.5: \; C_D = \frac{64}{\pi Re}(1 + 10^x) \tag{15a}$$

where $x = -0.883 + 0.906\,(\log Re) - 0.025(\log Re)^2$.

$$\text{For } 1.5 \leq Re \leq 135: \; C_D = \frac{64}{\pi Re}(1 + 0.138Re^{0.792}) \tag{15b}$$

Since the validity of Eq. (15) has been tested by several investigators using experimental results from various sources, for example, see [14, 15, 22], etc., it is assumed here to represent adequately the currently available experimental results in Newtonian fluids. The present drag values are in excellent agreement with the predictions of Eq. (15) within (±1.8%) and the other numerical results available in the literature [14–16, 18]. It is interesting to note here that though the results of Shenoy and Kleinstreur [18] correspond to $t/d = 0.1$, these are also in excellent agreement with the present and the other values [15, 16] which relate to $t/d = 0.05$, thereby suggesting only a weak effect of the disk thickness. Furthermore, since the experimental values of the drag on a disk in Newtonian fluids reported in [22] are within ~2% of Eq. (15), these are not repeated here. Furthermore, suffice it to add here that the present results on the wake length are within 2.4% of that reported in [18]. Similarly, Jossic et al. [56, 57] have reported velocity data for thin disks translating in Bingham plastic carbopol gels in the creeping flow limit. Figure 2 shows the comparison of the present numerical predictions (for three values of $t/d = 0.05$, 0.1, 0.15) with their experimental data for a range of values of t/d (0.02–0.16), as reported by Jossic et al. [56, 57]. Notwithstanding the scatter in their experimental results, the present numerical predictions seem to overestimate their values by up to 10% which is comparable to the experimental uncertainty of their data. Furthermore, the present predictions of the effect of disk thickness, though small, are qualitatively in line with the experimental results in so far that the drag coefficient increases slightly with the rising value of (t/d). Similarly, some experimental results on disk drag have been reported by Pazwash [53] and Pazwash and Robertson [54] which really relate to the values of the Reynolds number as high as $\sim 10^4$, but three data points relate to $Re < \sim 100$ and these values are in excellent agreement with the present predictions (Table 1). Note that the differences (of the order of ~10% or so) seen in Table 1 are well within the experimental errors in their data. Finally, following the reasoning of Overbeck [9], Table 2 compares the present values of the drag and Nusselt number with an oblate of aspect ratio of 0.1 in Newtonian fluids from the literature [75]. Considering the nature of the approximation involved here, the two heat transfer results differ from each other at most by 7% while the two drag values are within ~5% of each other.

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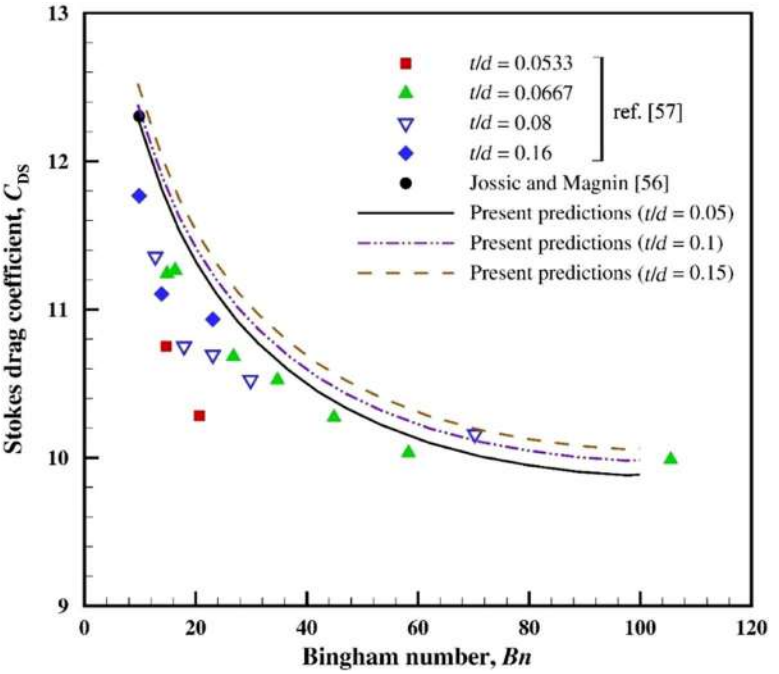


Figure 2. Comparison of drag coefficient with the results of Jossic and Magnin [56] and Jossic et al. [57] in the creeping limit.

Table 1. Comparison of present drag results with the experimental results of Pazwash [53] and Pazwash and Robertson [54].

Aqueous clay suspension (20%)			
Re	*Bn*	C_D[53, 54]	C_D (present)
102.2	4.7	5.82	4.984
76.5	25.3	14.5	13.24
45.3	10.7	22.7	20.89

Finally, scant experimental data on forced convection heat transfer from an isothermal disk in air are available in the literature [28]. While bulk of their results are for $Re > 100$, the present predictions are about ~20% lower than their experimental values in the range of ($30 \leq Re \leq 100$). These differences might be due to the wall effects and free convection effects present in the experimental results or due to the variation in thermo-physical properties, all of which are neglected in the present study. The preceding comparisons coupled with the detailed validations reported in some of our recent studies [41, 45–50] lend credibility to the present numerical results. Overall, these are considered to be reliable to within 2–3%.

The results for the creeping flow regime and that at finite Reynolds numbers are presented and discussed separately in the following sections.

Table 2. Comparison of drag and average Nusselt number values with that of Juncu [75] for an oblate spheroid of aspect ratio = 0.1 in Newtonian fluids.

	Reference [75]				Present			
			Nu				Nu	
Re	C_D	C_{DP}	Pr = 1	Pr = 10	C_D	C_{DP}	Pr = 1	Pr = 10
10	3.768	3.276	4.16	6.95	3.87	3.32	3.93	6.58
50	1.645	1.492	6.72	14.38	1.68	1.55	6.38	13.58
100	1.234	1.148	9.23	20.65	1.27	1.21	8.73	19.28

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4.2. Creeping flow regime

In the limit of $Re \longrightarrow 0$, the fluid inertia is negligible, but the momentum equation is still nonlinear owing to the Bingham constitutive equation. Since one of the distinct features of the flow of viscoplastic fluids is that such a material does not yield everywhere in the flow domain, Figure 3 shows the yield surfaces for scores of values of the Bingham number. As expected, there is a blob of unsheared material adhering at the front and rear stagnation points. At low Bingham numbers, this is in the form of small polar caps at the stagnation points. This region, however, grows both axially and radially with the increasing Bingham number eventually attaining a limiting shape in the form of a cone with its height of the order of $\sim 0.15d$. Further increase in Bn exerts only very weak influence. On the other hand, the fluid-like cavity encapsulating the disk is like a sphere flattened at the top and bottom extending up to $\sim 4d$ in both directions at $Bn = 0.2$ which eventually shrinks to about ~ 1.5–$2d$ in the radial direction and to a region of the order of d in the axial direction. It changes very little with further increase in the value of the Bingham number, thereby suggesting it to correspond to the fully plastic limit.

Using the standard definition of the drag coefficient C_D, one would expect it to increase with the Bingham number which is well approximated by the expression:

$$C_D \cdot Re = \frac{64}{\pi}(1 + 1.2Bn + 1.745Bn^{0.43}) \tag{16}$$

In the limit of $Bn \longrightarrow 0$, this expression reduces to the analytical expression due to Overbeck [9], that is, $C_D = (64/\pi Re)$. On the other hand, the drag force normalized by the yield stress force ($\tau_0 \cdot \pi R^2$) approaches a limiting value of ~ 12.7 consistent with the fully plastic flow regime ($Bn \to \infty$). The Stokes drag force normalized (using the yield stress multiplied by the projected area of the disk) was also plotted against the modified Reynolds number, Re^*. In the limit of $Re^* \to 0$, the drag force was seen to approach the limiting value of 12.7 (corresponding to fully plastic limit, $Bn \longrightarrow \infty$) and it rose steeply beyond $Re^* \sim 5 \times 10^{-4}$, and eventually attaining a very high value ($\sim 10^3$) at $Re^* = 10^{-3}$. This trend is also consistent with that seen for a sphere [39]. Finally, it is also useful to examine the scaling of the Nusselt number in the limit of the zero Reynolds number and low Peclet number. For instance, Brenner [76] proposed the following relationship for a particle of

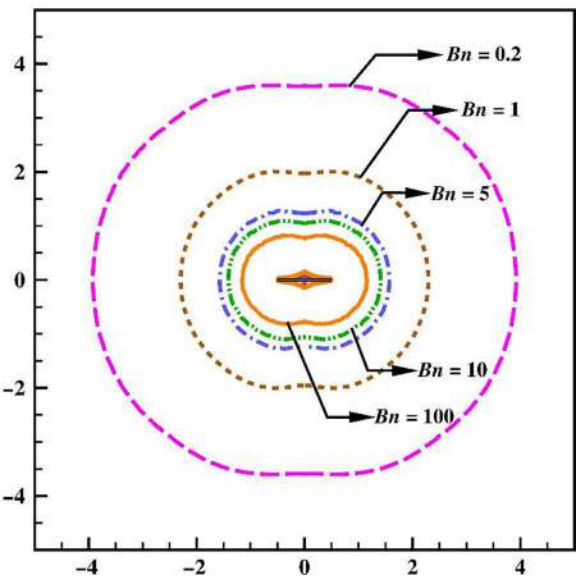


Figure 3. Structure of yield surfaces in the creeping flow limit. Flow is in the upward direction.

arbitrary shape submerged in Newtonian fluids:

$$\frac{Nu}{Nu_\infty} \sim 1 + \frac{Nu_\infty}{8} Pe \tag{17}$$

where Nu_∞ is the limiting value of the Nusselt number in the conduction regime (obtained here by the solution of the conduction equation for the two boundary conditions, i.e., CWT and CHF as $Nu_\infty = 2.48$ and $Nu_\infty = 2.34$, respectively, which are in line with the upper and lower bound estimates available in the literature [77]. Indeed, the present results conform to the scaling suggested by Eq. (17). It is virtually impossible to discern any effect of Bingham number although the results for Newtonian ($Bn = 0$) and viscoplastic fluids begin to differ from each other at about $Pe = 0.03$ and $Pe = 0.05$ for the CWT and CHF conditions, respectively. This is perhaps not surprising because the Peclet number is really independent of the viscosity of the fluid and therefore even the use of Re^* and Pr^* will not change this relationships.

4.3. Finite Reynolds number regime

4.3.1. Hydrodynamic characteristics

As the Reynolds number is progressively increased, additional features begin to appear in the velocity and temperature fields, which directly influence the values of the total drag coefficient and Nusselt number. We begin with a discussion of the morphology of the yield surfaces in the next section.

A. Yield surfaces: It is useful to recall here that the fluid inertia (Re) tends to promote fluid yielding, whereas this propensity is suppressed by the yield stress (Bn) effects. Figure 4 shows this interplay between these two factors, thereby leading to a range of possibilities. First, there is a good agreement between the predictions of all three regularization schemes used here. The minor differences seen in the remote boundary within ±10–15° of the vertical centerline are not of great significance, for the values of both C_D and Nu are primarily determined by the gradients on the surface of the disk. As postulated above, at low Bingham numbers, such as $Bn = 1$ and $Bn = 5$, the yielded region is seen to grow in the downstream direction due to decaying stress levels, whereas there is a little change in the lateral direction. Also, the unyielded material adhering at the front stagnation point diminishes with the increasing Reynolds number due to high deformation rates in this region, but it grows with the Bingham number. On the other hand, the unyielded region in the rear of the disk grows with increasing Re or Bn or both. From another perspective, with increasing inertial effects (i.e., increasing Reynolds number), there is a growing tendency for an adverse pressure gradient to be established in the diverging flow area in the rear of the disk which can lead to the formation of the recirculating pattern. This is known to occur at about $Re = 0.6$ in Newtonian fluids ($Bn = 0$). Due to the augmented viscous effects in the present case, this behavior is suppressed in Bingham fluids. It thus stands to reason that for a given Reynolds number, there must be a limiting Bingham number, beyond which no recirculating region (wake) is formed, or conversely, for a given Bingham number, there must be a limiting Reynolds number required for the wake formation to occur. These critical points were located by a careful scrutiny of the surface vorticity and the velocity vector plots in the rear of the disk. Based on the examination of the streamline and velocity vectors, the wake lengths were determined as functions of the Reynolds and Bingham numbers. This procedure was also used to ascertain the critical values of the Reynolds number (Re_{cr}) and Bingham (Bn_{cr}) numbers. The results plotted in Figure 5*a, b* show the variation of the wake length with Bn and Re, and the functional relationship between the Re–Bn_{cr} or Bn–Re_{cr} delineating the wake/no wake domains, respectively. First, the positive dependence of the wake length on the Reynolds number is seen not only to gradually weaken with the increasing Bingham number, but the wake length is also dramatically shortened, almost by a factor of 3 or so at $Re = 100$. This trend is qualitatively consistent with that observed for other axisymmetric shapes also [41, 46, 48]. Figure 5*b* shows that, as expected, the value of the Bingham number required to completely suppress the wake formation increases with the increasing Reynolds number, that is, wake will only form below this line. Conversely, the critical Reynolds number (above

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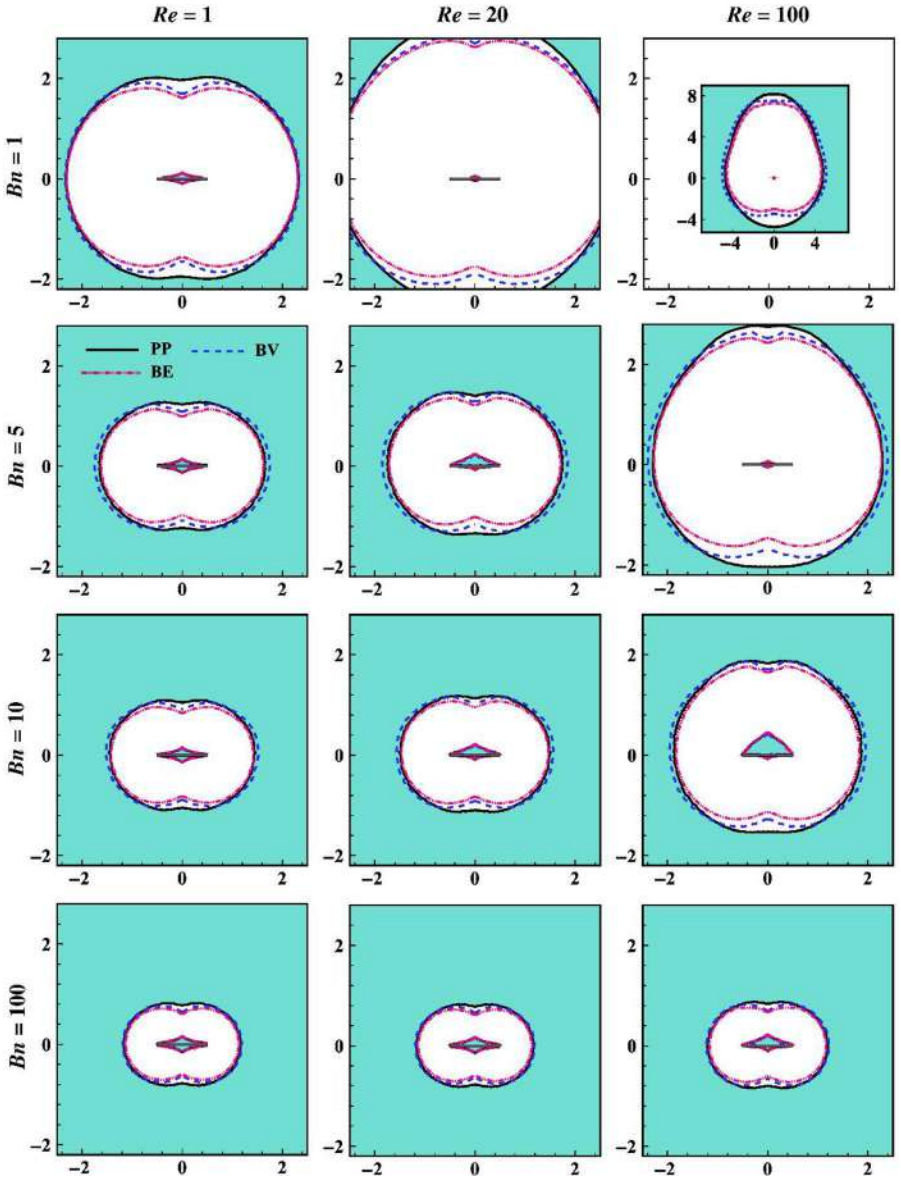


Figure 4. Effect of the Reynolds and Bingham numbers on the structure of yielded and unyielded regions predicted by different regularization schemes. PP, Papanastasiou model; BE, Bercovier and Engleman model; BV, bi-viscosity model.

which recirculating region will form in the rear) also increases with the Bingham number, that is, wake will only form above this line.

The present numerical values of the critical Reynolds and Bingham numbers are well fitted by the following expression:

$$Re_{cr} = 0.6 + 0.26Bn^{1.28} \tag{18}$$

$$Bn_{cr} = 6.49 + 1.77Re^{0.897} \tag{19}$$

The average and maximum deviations associated with Eq. (18) are 2.8 and 5.9% and that for Eq. (19) are 3.7 and 6.5%, respectively.

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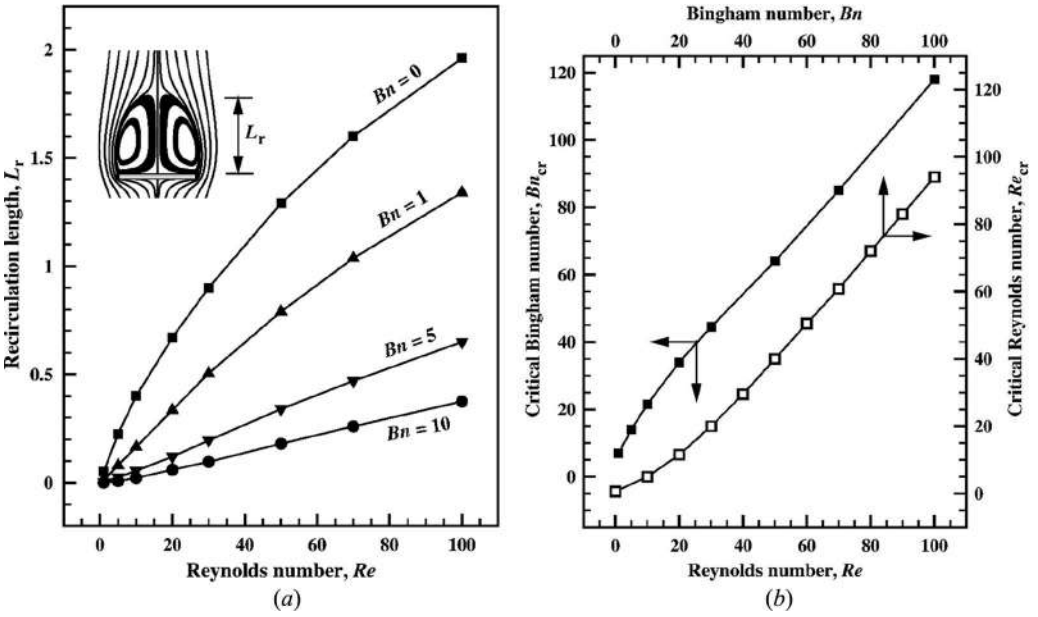


Figure 5. (*a*) Dependence of recirculation length on the Reynolds and Bingham number. (*b*) Critical values of the Bingham and Reynolds number for the suppression and onset of flow separation, respectively.

B. Drag coefficient: The overall hydrodynamic drag experienced by the disk is made up of two components, namely, the form or pressure component due to the normal stresses (acting on the flat surfaces) and the viscous or friction drag due to the tangential stresses (acting on the lateral surface with negligible surface area of $\pi\, d \cdot t$). It is thus reasonable to postulate that the form drag should dominate the total drag. Indeed, this conjecture is well borne out by the results shown in Figure 6*a* where this ratio is plotted against the Reynolds number. At low Reynolds numbers, this ratio is constant but it takes off at about $Re = 20$, suggesting that the normal stresses rise much more steeply than the tangential stresses. Following the arguments used in previous studies [41, 46, 48], the definition of the Reynolds number (Re^*), given by Eq. (11) which takes into account the role of fluid yield stress, is more appropriate for correlating the data in such fluids.

Indeed, this idea of combining Re and Bn into Re^* was initially proposed more than 50 years ago by Ansley and Smith [78] for correlating their sphere drag results in Bingham plastic fluids and it has been subsequently used by others, for example, see [41, 46, 48–50]. At the outset, the present drag results are plotted against Re^*, Figure 6*b,* along with Eq. (15). The discrepancies between the two can then be ascribed to the Bingham plastic behavior. Indeed, the present results do follow Eq. (15) except for the fact that it consistently under-predicts the present numerical values, especially at low values of Re^*.

Based on ~80 numerical data points for $t/d = 0.05$, the following expression correlates the present drag coefficient results slightly better than Eq. (15):

$$C_D = \frac{64}{\pi Re^*}\left(1 + a_1 Re^{*b_1}\right) \tag{20}$$

The values of the fitting constants are as follows:

For $Bn = 0$, $a_1 = 0.138$, $b_1 = 0.793$, and for $0 < Bn \leq 100$, $a_1 = 0.603$, $b_1 = 0.138$. For Newtonian fluids ($Bn = 0$), the mean and maximum deviations are 1.9 and 2.4%, respectively. The corresponding figures for $0 < Bn \leq 100$ are 11.8 and 28.5%, respectively.

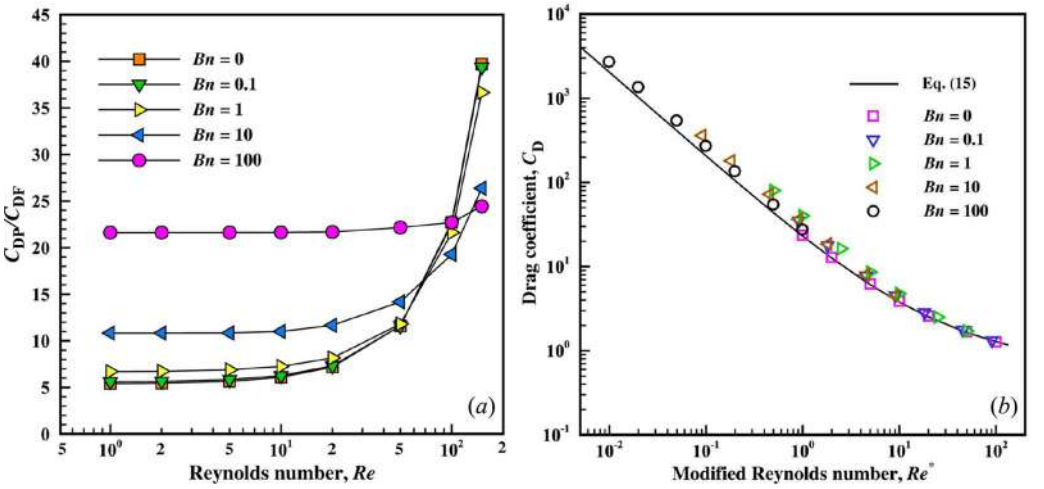


Figure 6. Dependence of (*a*) pressure to friction drag ratio on the Reynolds and Bingham numbers; (*b*) consolidation of the effect of Bingham numbers for C_D vs. Re^* plot.

4.3.2. Heat transfer aspects

A. Isotherm contours: Figure 7 shows the representative isotherm contours for a range of conditions for the CWT boundary condition on the disk surface. At low Peclet numbers, advection is weak and heat transfer occurs primarily by diffusion as is evident by the more or less symmetric isotherms (top row and left half of Figure 7). This trend is further accentuated by the increasing fluid yield stress.

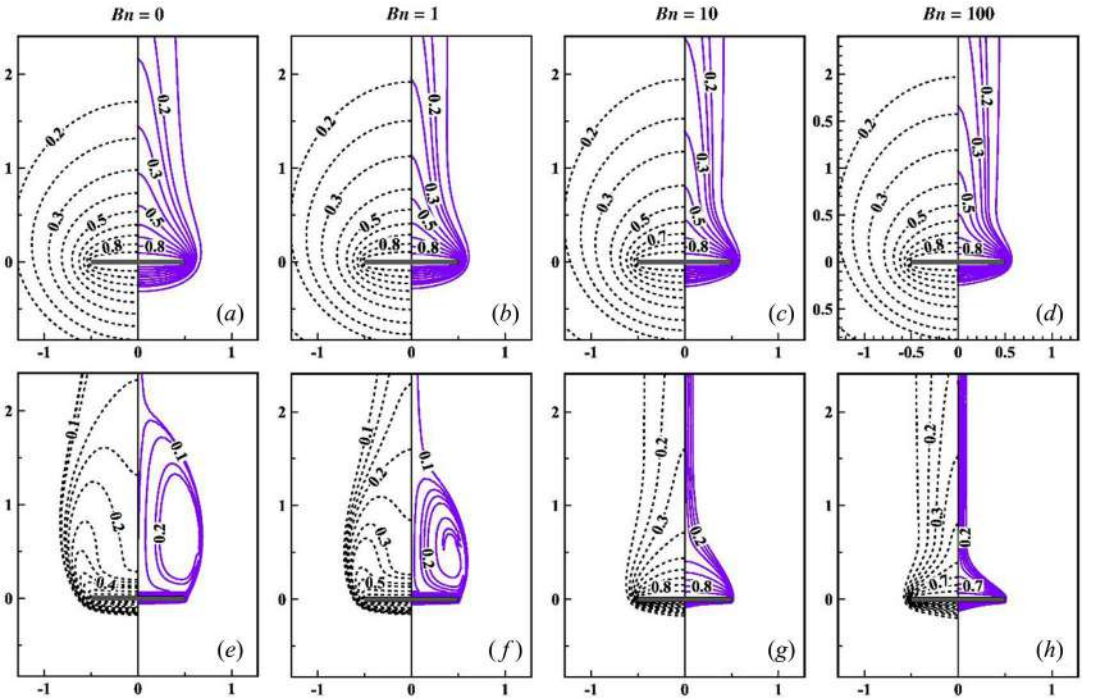


Figure 7. Representative isotherm contours for CWT condition at $Pr = 1$ (left half) and $Pr = 100$ (right half). (*a–d*) $Re = 1$; (*e–h*) $Re = 100$. *Note*: CWT, constant wall temperature.

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However, at the same value of $Re = 1$, the isotherms shown in the right half of the top row in Figure 7 exhibit distortion and stretching downstream from the disk as the value of the Peclet number has increased by a factor of 100. The increasing degree of clustering of isotherms in the front indicates the sharpening of the temperature gradient in this region.

At $Re = 100$ (bottom row, Figure 7), no new features are observed except for the fact that further thinning of the thermal boundary layer occurs. Based on these observations, it is perhaps fair to postulate that heat transfer should bear a positive dependence on the both Reynolds and Prandtl numbers for a fixed value of *Bn*. This is in line with the classical boundary layer approximation. However, the role of Bingham number is more complex, in so far that with the increasing Bingham number, the fluid-like layer close to the disk becomes very thin, thereby sharpening the temperature gradient. This should facilitate heat transfer. On the other hand, the expanse of the unsheared material adhering in the rear of the disk, as seen in Figure 4, adds to the thermal resistance, thereby lowering the rate of heat transfer. From another perspective, as the tendency for wake formation is also suppressed and this further leads to some reduction in heat transfer. Indeed, all these conjectures are borne out by the local and average Nusselt number results presented in the ensuing sections. Finally, suffice it to add here that the analogous isotherm patterns for the CHF condition are qualitatively similar to that seen in Figure 7 whence these are not included here for the sake of brevity.

B. Local Nusselt number: Depending upon the thermal boundary condition prescribed on the surface of the disk, the local Nusselt number is either determined by the value of the temperature gradient normal to the surface (for the CWT condition) or by the temperature of the surface of the disk (for the CHF condition). The fluid properties indirectly influence these results via influencing the velocity field which, in turn, determines the temperature field. Thus, the velocity gradient (second invariant of the rate of strain tensor) on the surface of the disk also varies from one point to another, for example, it is maximum at the front stagnation point. This is of no consequence for a Newtonian fluid because their viscosity is independent of shear rate. However, it is not so in the case of Bingham plastic fluids, thereby enhancing the role of viscosity and yield stress. Therefore, the value of the local Nusselt number at a point is governed by these two effects both of which vary differently along the surface of the disk.

Figure 8 shows the representative results on the variation of the local Nusselt number along the surface of the disk for a range of conditions. Owing to the axisymmetry, these profiles correspond to only half-disk. For the CWT condition, Figure 8*a,* the positive roles of *Re* and *Pr* are seen to be in line with what is postulated in the preceding section. Irrespective of the boundary condition (CWT or CHF), the Nusselt number is seen to be minimum at the front stagnation point which is due to the polar caps of unsheared material present here. It increases slightly as a fluid element travels along the line AB and the two peaks seen here correspond to the two corners B and C where the isotherms and streamlines must bend to negotiate the body contour and thereafter the local Nusselt number continues to decrease all the way up to point D. This behavior is seen in the conduction-dominated regime (low *Pe*/high *Bn*). As the Reynolds number is gradually increased, the Nusselt number shows some recovery in the rear beyond the point of separation if a wake is present. On the other hand, for fixed values of *Re* and *Pr,* varying levels of enhancement are evident in heat transfer with the increasing value of the Bingham number. However, the effect of Prandtl number is progressively enhanced on the upstream surface and suppressed in the rear due to the expanding unyielded material present with the increasing Bingham number. Depending upon the relative magnitudes of these two effects, the overall average Nusselt number may increase or decrease, as will be seen in the next section.

Now turning our attention to Figure 8*b,* while the effects of the Prandtl and Reynolds numbers are qualitatively similar to that seen in Figure 8*a,* the overall values of the heat transfer coefficient are generally lower in the CHF case than that in CWT case. This is presumably so due to the fact that the CHF demands the product of $h(T - T_0)$ to remain constant at each point on the surface of the disk. Due to the formation of the solid-like regions (which offer high thermal resistance), high temperature gradient is needed to achieve the same heat flux which translates into smaller values of the Nusselt number.

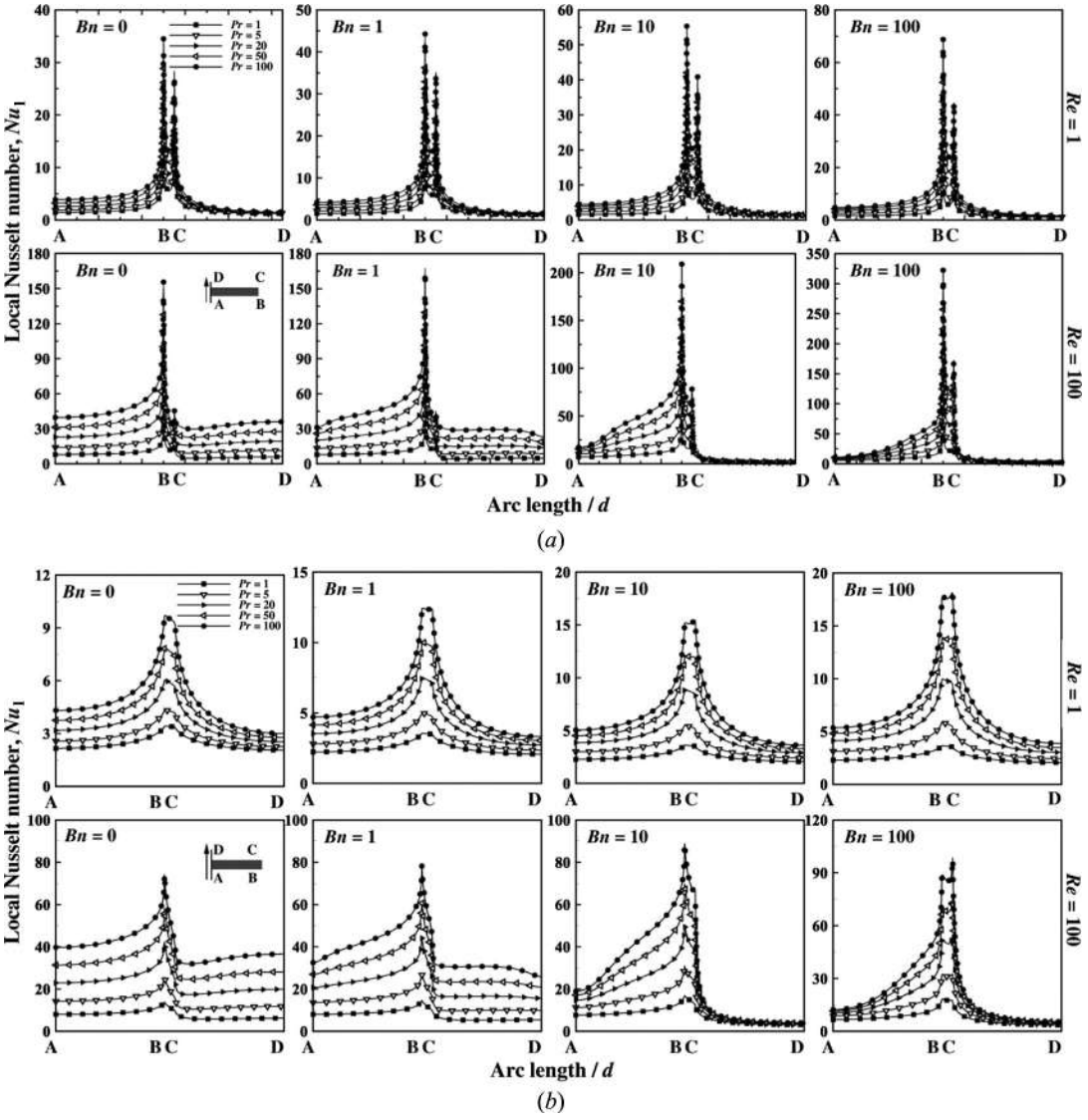


Figure 8. (*a*) Variation of local Nusselt number along the surface of disk for CWT condition. (*b*) Variation of local Nusselt number along the surface of disk for CHF condition. *Note*: CHF, constant heat flux; CWT, constant wall temperature

C. Average Nusselt number: The overall heat transfer behavior is typically described in terms of the surface averaged Nusselt number. Dimensional considerations suggest it to be a function of *Re, Pr, Bn,* and (t/d) or of Re^*, Pr^*, and (t/d), for a given thermal boundary condition. For a fixed value of $(t/d = 0.05)$ and thermal boundary condition, the functional dependence between *Nu, Pr, Re,* and *Bn* is exemplified by the results shown in Figure 9 for the two boundary conditions used here. The positive influence of *Re* and *Pr* on the average Nusselt number need not be overemphasized here. Furthermore, the fact that Nu_H is consistently lower than Nu_T, is also in line with the explanation offered in the previous section. However, the role of the Bingham number is clearly modulated by the value of the Reynolds number irrespective of the value of the Prandtl number. Broadly, at low Reynolds numbers, yield stress effects dominate and therefore, there is very little influence of the thermal boundary condition. However, this difference between Nu_H and Nu_T is magnified with the increasing extent of convection at low Bingham numbers due to the increasing value of the Peclet

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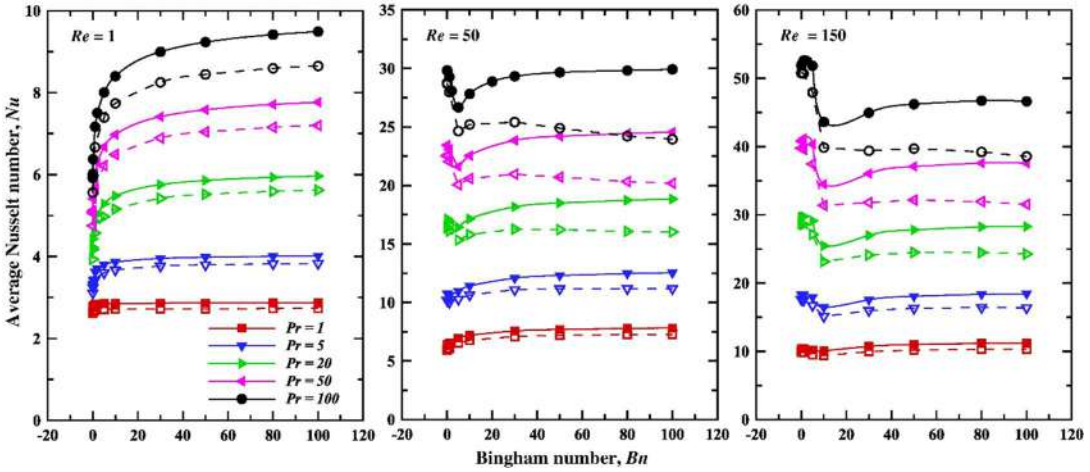


Figure 9. Dependence of average Nusselt number on the Reynolds number, Prandtl number, and Bingham number (solid lines: Nu_T; dashed lines Nu_H).

number. On the other hand, all else being equal, the average Nusselt number is seen to attain a limiting value with respect to the Bingham number. The 100-fold increase in the Peclet number is seen to increase the value of the average Nusselt number by a factor of ~3 which suggests the scaling of $Nu \sim Pe^{1/4}$ which is slightly lower than the usual scaling of $Nu \sim Pe^{1/3}$ seen in Newtonian fluids. This difference may be due to the formation of the polar caps of the unyielded material on the surface of the disk. On the other hand, at $Re = 100$, while the overall trends at $Bn \geq 30$ are similar except for the fact that the Nusselt number is seen to scale as $Nu \sim Pe^{1/3}$. This suggests the extended viscous-like regime due to the fluid yield stress which resists inertial effects, that is, it behaves like a viscous fluid. There is a slight decrease in the Nusselt number from its maximum value in Newtonian fluids by the introduction of yield stress, and it exhibits a weak minimum before attaining a constant value independent of the Bingham number. The maximum value at $Bn = 0$ is simply due to the fact that there is no unyielded material present on the surface of the disk. At $Re = 150$, the polar cap in the rear expands which leads to the reduction in heat transfer and eventually it attains a limiting size and hereafter, the effect of Bn weakens. It needs to be recognized here that the overall heat transfer is determined by a complex interaction between three boundary layers: momentum, thermal, and stress. Generally, the thermal boundary layer thickness is smaller than the momentum boundary layer thickness for $Pr > 1$. However, depending upon the extent of the stress boundary layer, the fluid inside the thermal boundary layer may be completely or partially yielded. Suffice it to add here that the three boundary layers scale differentially with *Re*, *Pr,* and *Bn*. Therefore, the minima seen in Figure 9 at $Re = 50$ and $Re = 150$, especially at high Prandtl numbers corresponds to when the bulk of the fluid inside the thermal boundary layer is unyielded, thereby leading to the minimum heat transfer. Following the highly successful approach of correlating such data in previous studies [41, 46, 48–50], the present numerical results are consolidated in terms of the Colburn heat transfer factor, j_H, as:

$$j_H = \frac{Nu}{Re^* \cdot Pr^{*1/3}} = a_2 (Re^*)^{-2/3} \tag{21}$$

Equation (21) reproduces ~500 individual data points for each thermal boundary condition. For the CWT condition corresponding values of constant a_2 for $Re^* < 1$ and $1 \leq Re^* \leq 150$ are 2.21 and 1.70 with associated mean deviations of 15% and 11% which rise to their maximum values of 38% and 39%, respectively. For the CHF condition, the corresponding values of constant a_2 for $Re^* < 1$ and $1 \leq Re^* \leq 150$ are 2.05 and 1.74 with the mean deviations of 10% and 11% and the maximum deviations of 33% and 37%, respectively. It must be added here that, however, only about ~8% data exhibit

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deviations greater than 25% without any discernible trends. Some observations about the form of Eq. (21) are in order here. First, it includes the limiting case of Newtonian fluid behavior, that is, $Bn = 0$ when Re^* and Pr^* correctly reduce to their expected forms for Newtonian fluids. Second, it also retains the widely accepted scaling of ($Nu \sim Pr^{1/3}$) in line with the classical boundary layer analysis. The power of (−2/3) for Re^* in Eq. (21) is also consistent with its widespread use in the literature in the context of Newtonian fluids.

4.4. Effect of disk thickness

The effect of disk thickness on the drag and Nusselt number stems from two considerations. First, as the value of (t/d) increases, it becomes more like a short cylinder and therefore, strictly speaking the velocity and temperature fields are likely to be three-dimensional due to the end effects. Second, if one can still approximate it as an axisymmetric flow, the lateral surface area increases and therefore the viscous drag must increase, thereby leading to the higher drag than that for a disk of zero thickness ($t/d \to 0$). For Newtonian fluids, this increase is of the order of ~3.5% as the value of (t/d) is gradually increased from 0.01 to 0.075 in the creeping flow regime. This compares favorably with the experimental results of Ui et al. [74] which predicts an increase of ~3% over this range of (t/d). As the Reynolds number is gradually incremented, this effect reduces from 3.5% at $Re = 1$ to about 0.5% at $Re = 100$. In Bingham plastic fluids, the contribution of wake drag is reduced but the yield stress makes a positive contribution to the total drag. Thus, depending upon the values of Bn and Re or Re^*, the total drag may slightly increase or decrease with the increasing value of (t/d). However, the variation is seldom greater than 4.5–5% (Table 3) depending upon whether the wake is formed or not in a given situation.

Similarly, the value of the Nusselt number (Table 4) is seen to be influenced in a complex manner by Bn, Re, and Pr. For Newtonian fluids ($Bn = 0$), the effect of (t/d) is only minor in every case. This, however, gets accentuated with the increasing Bingham number leading to almost 15% increase in the value of the average Nusselt number in extreme cases. Whenever a wake is present (e.g., at $Bn = 1$ and $Re = 100$), the average Nusselt number shows decreasing value with the increasing disk thickness. In the absence of a wake, it shows a positive dependence on (t/d) which is presumably purely a geometric effect. Thus, the effect of (t/d) on drag and Nusselt number is entwined with the formation and suppression of wake and/or the size and location of the unyielded regions which, in turn, are determined by the values of Re and Bn.

Before leaving this section, it is appropriate to reflect on the implications of the key assumptions inherent in this work. Owing to the relatively small values of $\Delta T = T_w - T_0 \leq 5\,\mathrm{K}$ or the surface flux ($1\,\mathrm{W\,m^{-2}}$) used here, the temperature dependence of the thermophysical properties has been

Table 3. Effect of disk thickness (t/d) on total drag coefficient for a range of Bingham and Reynolds numbers.

t/d	0.01	0.025	0.05	0.075	0.01	0.025	0.05	0.075
Re	$Bn = 0$				$Bn = 1$			
1	23.070	23.267	23.591	23.906	77.728	78.622	80.074	81.475
2	12.572	12.681	12.862	13.038	38.972	39.418	40.142	40.843
5	6.0901	6.141	6.2256	6.309	15.872	16.048	16.333	16.611
10	3.8031	3.8291	3.8736	3.9183	8.3410	8.4250	8.5607	8.6942
50	1.6893	1.6874	1.6861	1.6864	2.5029	2.5017	2.5018	2.5076
100	1.2772	1.2751	1.2720	1.2695	1.7344	1.7276	1.7145	1.7059
150	1.0774	1.0761	1.0743	1.0733	1.4350	1.4279	1.4161	1.4080
Re	$Bn = 10$				$Bn = 100$			
1	350.23	354.68	361.63	368.63	2638.0	2669.9	2721.3	2772.8
2	175.16	177.38	180.86	184.35	1319.0	1335.0	1360.7	1386.4
5	70.184	71.068	72.450	73.841	527.64	534.02	544.29	554.59
10	35.302	35.732	36.410	37.093	263.87	267.05	272.18	277.33
50	7.8409	7.8934	7.9882	8.0848	53.098	53.696	54.681	55.690
100	4.4540	4.4658	4.4895	4.5133	26.996	27.255	27.697	28.172
150	3.3500	3.3222	3.2997	3.3035	18.398	18.537	18.800	19.095

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Table 4. Effect of disk thickness (t/d) on average Nusselt number at $Pr = 100$ for a range of the Bingham and Reynolds numbers.

		Nu_T				Nu_H			
	Re	$t/d = 0.01$	$t/d = 0.025$	$t/d = 0.05$	$t/d = 0.075$	$t/d = 0.01$	$t/d = 0.025$	$t/d = 0.05$	$t/d = 0.075$
$Bn = 0$	1	5.7435	5.8328	5.9144	5.9159	5.3250	5.4299	5.5610	5.6501
	2	6.8598	7.0023	7.1397	7.1686	6.3376	6.5042	6.7114	6.8530
	5	8.9497	9.1962	9.4438	9.5286	8.1909	8.4899	8.8620	9.1185
	10	11.758	11.980	12.210	12.274	10.837	11.096	11.470	11.752
	50	30.658	30.388	29.837	29.167	28.728	28.729	28.747	28.625
	100	44.392	44.031	43.075	41.750	41.873	41.741	41.810	41.249
	150	53.697	53.007	51.855	50.232	51.020	50.792	50.757	49.949
$Bn = 1$	1	6.8533	7.0105	7.1689	7.2129	6.2801	6.4519	6.6682	6.8192
	2	8.0967	8.3316	8.5730	8.6604	7.3768	7.6264	7.9369	8.1535
	5	10.260	10.632	11.018	11.181	9.3324	9.7162	10.191	10.523
	10	12.645	13.126	13.624	13.843	11.612	12.105	12.722	13.156
	50	30.311	29.876	29.272	28.551	28.050	27.952	27.974	27.868
	100	44.866	44.460	43.175	41.872	41.577	41.361	41.213	40.720
	150	54.825	54.076	52.589	50.581	51.323	50.830	50.687	49.610
$Bn = 10$	1	7.9660	8.1867	8.4020	8.4736	7.2416	7.4669	7.7381	7.9278
	2	9.4740	9.8105	10.149	10.288	8.5212	8.8448	9.2326	9.5065
	5	11.955	12.514	13.081	13.343	10.595	11.098	11.700	12.126
	10	14.428	15.197	15.978	16.353	12.683	13.355	14.170	14.742
	50	25.689	26.697	27.820	28.305	22.683	23.737	25.205	26.164
	100	34.679	35.087	36.087	36.381	29.838	30.980	32.781	33.833
	150	52.474	49.712	43.601	41.884	45.903	44.203	39.845	39.169
$Bn = 100$	1	9.0831	9.3029	9.4908	9.5268	8.2057	8.4196	8.6517	8.8082
	2	10.938	11.286	11.614	11.731	9.7262	10.028	10.368	10.609
	5	13.931	14.553	15.177	15.454	12.093	12.565	13.115	13.519
	10	16.693	17.626	18.559	18.996	14.192	14.848	15.620	16.189
	50	26.278	28.129	29.919	30.766	21.284	22.520	23.944	24.990
	100	34.839	36.911	38.947	39.752	28.223	29.811	31.706	32.908
	150	42.307	44.360	46.610	47.397	34.236	36.028	38.547	40.045

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neglected. Among all the physical properties, perhaps the yield stress and Bingham plastic viscosity are expected to be the most temperature sensitive properties. Unfortunately, not only such data is scarce in the literature but often conflicting findings have been reported in the literature concerning the effect of temperature on the yield stress [79, 80]. Thus, for instance, for the commonly used model Bingham fluids (carbopol solutions), the yield stress was reported to remain nearly constant over the temperature range of 10–85°C [79, 80]. In contrast, Weber et al. [81] report a non-monotonic variation of the yield stress with temperature for nominally the same fluid as that used in [79, 80]. Therefore, it is not yet possible to postulate reliable forms of temperature-yield stress/Bingham viscosity behavior. In view of this uncertainty, the assumption of the constant properties seems reasonable, at least over the narrow window of the temperature range used here. Second, the viscous dissipation effects have been neglected here by arguing that the value of the Brinkman number (based on μ_B) is very small over the range of conditions spanned here. While these two assumptions lead to the decoupling of the velocity and temperature fields in the present case, one needs to establish the validity of these assumptions in each case by establishing the severity of the temperature dependence of the Bingham fluid properties.

5. Conclusion

In this work, extensive numerical results on the momentum and heat transfer phenomena from an isothermal or isoflux circular disk in transverse flow of Bingham plastic fluids have been studied over wide ranges of parameters as $Re \leq 150$, $Bn \leq 100$, and $1 \leq Pr \leq 100$ and for four different values of disk thickness ($t/d = 0.01$, 0025, 0.05, and 0.075) to delineate the effect of each of these parameters on the overall drag and heat transfer although the bulk of results relate to $t/d = 0.05$. While most of the results presented in this work are based on Papanastasiou–Bingham regularization model, limited results are also included here using the bi-viscosity and Bercovier and Engelman approximations

to substantiate the reliability and precision of the present results. Broadly, the yielded regions expand with the increasing Reynolds number while the Bingham number tends to suppress this tendency. Due to the enhanced viscous effects, the propensity for flow separation is reduced in Bingham media. The critical (minimum) Reynolds number for the flow to detach itself increases with the increasing yield stress effects. Similarly, the higher the Reynolds number (fluid inertia), larger is the Bingham number (Bn_{cr}) needed to stop the flow detachment. Other than this, the disk experiences higher drag in Bingham plastic fluids than that in Newtonian fluids ($Bn = 0$), in line with the other axisymmetric shapes. Under most conditions, the form drag dominates the total drag, contributing up to 90% to the total drag on the disk. Similarly, the heat transfer characteristics are influenced by the values of *Bn, Re, Pr,* and (t/d) in a complex manner. While the increasing values of *Bn* suppress fluid yielding thereby sharpening the temperature gradient which favors heat transfer, some of this gain is offset by the formation of unsheared blob of material adhering to the disk surface. The effect of disk thickness is much more severe on the Nusselt number (~15%) than that on drag (~5%). Finally, simple predictive expressions are developed as functions of the modified Reynolds (Re^*) and Prandtl (Pr^*) numbers which can be used to make *a priori* estimation of the settling velocity (or drag) and Nusselt number in a new application.

Acknowledgments

R. P. Chhabra would like to thank the Department of Science and Technology, Government of India, New Delhi for the award of a J. C. Bose fellowship to him for the period 2015–2020.

Funding

This work was supported by Department of Science and Technology, Ministry of Science and Technology, Government of India: [Grant number SB/S2/JCB-06/2014].

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